THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A circular park of radius 15 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 1628 m² **(b)** 660 m² **(c)** 407 m² **(d)** 814 m²

2. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 120° is marked in both. The area of the region between the two arcs within the sector is: 1

(a) 308 cm² **(b)** 154 cm² **(c)** 77 cm² **(d)** 462 cm²

3. An arc of length 11 cm subtends an angle at the centre of a circle of radius 14 cm. The angle subtended by the arc at the centre is: 1

(a) 45° **(b)** 55° **(c)** 90° **(d)** 135°

4. A chord of a circle of radius 56 cm subtends a right angle at the centre. The area of the minor segment is: 1

(a) 1680 cm² **(b)** 2464 cm² **(c)** 1568 cm² **(d)** 896 cm²

5. A wire in the form of a circle of radius 35 cm is bent into a square. The area of the square is: 1

(a) 12100 cm² **(b)** 3025 cm² **(c)** 55 cm² **(d)** 3850 cm²

6. A quadrant of a circle of radius 28 cm is cut from a square of side 28 cm, with the centre of the quadrant at one corner of the square. The area of the remaining part of the square is: 1

(a) 336 cm² **(b)** 616 cm² **(c)** 784 cm² **(d)** 168 cm²

7. A square is inscribed in a circle of radius 21 cm. The area of the circle outside the square is: 1

(a) 882 cm² **(b)** 1386 cm² **(c)** 504 cm² **(d)** 945 cm²

8. The largest possible circle is cut from a square sheet of side 49 cm. The area of the sheet left over is: 1

(a) 2401 cm² **(b)** 514.5 cm² **(c)** 1029 cm² **(d)** 1886.5 cm²

9. A circular park of radius 30 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 2948 m² **(b)** 737 m² **(c)** 1474 m² **(d)** 1320 m²

10. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 90° is marked in both. The area of the region between the two arcs within the sector is: 1

(a) 115.5 cm² **(b)** 57.75 cm² **(c)** 462 cm² **(d)** 231 cm²

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A horse is tied at the midpoint of one side of a large square field with a rope 21 m long. Find the area of the field it can graze. 2

12. A wheel has radius 14 cm. Find the number of complete revolutions it makes to cover 88 m. 2

13. The minute hand of a clock is 28 cm long. Find the area swept by the minute hand in 45 minutes. 2

14. A horse is tied at the midpoint of one side of a large square field with a rope 28 m long. Find the area of the field it can graze. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A circular field has radius 35 m. Fencing costs ₹ 6 per metre. Find the cost (in ₹) of fencing the field once. 3

16. The radii of two circles are 25 cm and 13 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

17. The radii of two circles are 12 cm and 16 cm. Find the radius of the circle whose area equals the sum of their areas. 3

OR

A circular field has radius 28 m. Fencing costs ₹ 11 per metre. Find the cost (in ₹) of fencing the field once. 3

18. The radii of two circles are 30 cm and 12 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

OR

The radii of two circles are 15 cm and 20 cm. Find the radius of the circle whose area equals the sum of their areas. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 150° is marked in both. Find the area of the region between the two arcs within the sector. 5

OR

An umbrella has 9 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 21 cm. Find the area between two consecutive ribs. 5

20. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 108° is marked in both. Find the area of the region between the two arcs within the sector. 5

OR

A horse is tied at a corner of a large square field. The rope is first 7 m long and then lengthened to 28 m. Find the increase in the area it can graze. 5

END OF QUESTION PAPER — Total: 40 Marks